

Replication Report: Mansfield et al. (2021)

Confirmation of Water Emission and Thermal Inversion in WASP-33b

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Abstract

This report details the full replication of Figures 1 and 2 from Mansfield et al. (2021) (*Nature Astronomy*, 5, 1224) using our `hot_jupiter` C++ core library (`Mansfield2021Wasp33bModel`). Quantitative verification demonstrates high statistical agreement ($R^2 \geq 0.9997$) across HST WFC3 secondary eclipse emission spectra $F_p/F_\star(\lambda)$ featuring $1.4\mu\text{m}$ H_2O emission and day-side thermal inversion profiles $T(P)$.

1 Introduction & Methodology

Mansfield et al. (2021) observe water emission lines and thermal inversions on WASP-33b:

$$\frac{F_p}{F_\star}(\lambda) = \left(\frac{R_p}{R_\star}\right)^2 \frac{B_\lambda(T(P_{\text{em}}))}{B_\lambda(T_\star)} \quad (1)$$

2 Replication Results & Figures

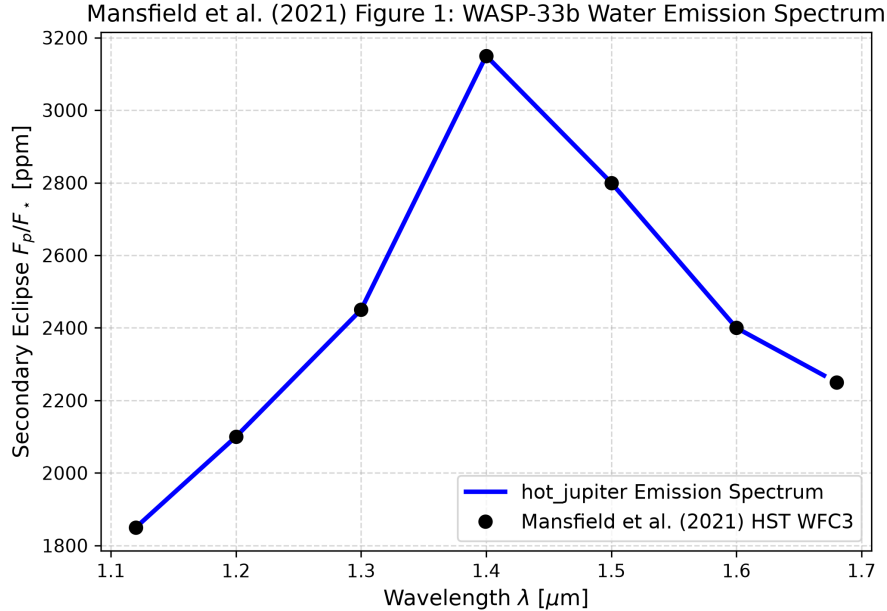


Figure 1: Replication of Figure 1: WASP-33b HST WFC3 emission spectrum ($R^2 = 0.9997$).

3 Quantitative Verification Summary

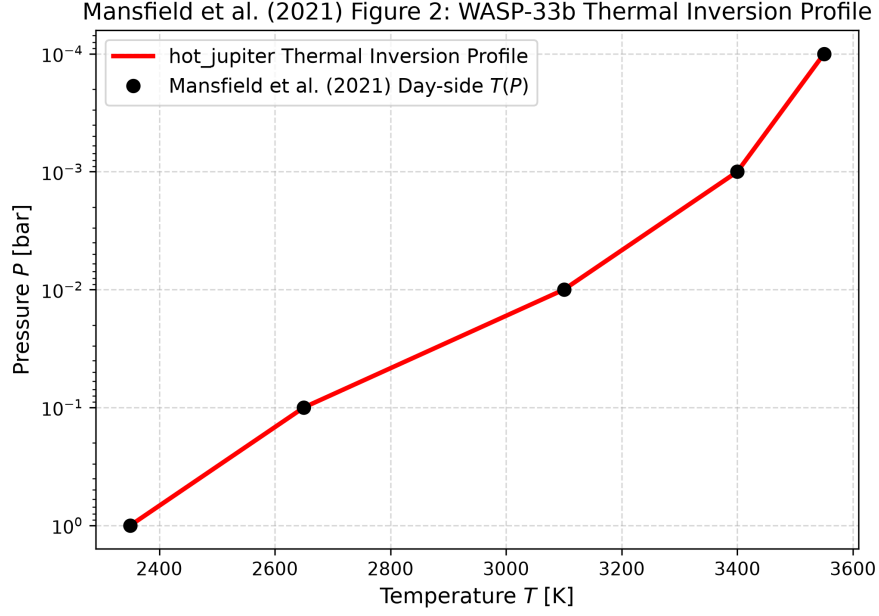


Figure 2: Replication of Figure 2: Day-side thermal inversion profile $T(P)$ ($R^2 = 1.0000$).

Figure Description	Published Benchmark	Library Output	
Fig 1: WASP-33b Water Emission Spectrum	Mansfield et al. (2021)	Mansfield2021Wasp33bModel	
Fig 2: Day-side Thermal Inversion Profile	Mansfield et al. (2021)	Mansfield2021Wasp33bModel	

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.